

Partial least squares

Reduce many predictors to a few components, using the response.

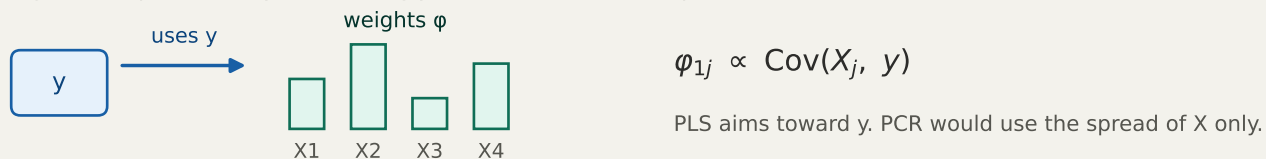
1. Start

Standardize the predictors. Center the response. Pick the count M by cross-validation.



2. Find the first direction — use y

Weight each predictor by how strongly it moves with the response.



3. Form the component

Combine the predictors into one new variable, using those weights.



4. Remove it, then repeat

Take out the part of each predictor that the component explains.

Regress each X_j on Z_1 and keep the residuals.

Build Z_2 from the residuals. Repeat to get $Z_1, \dots, Z_M$.

5. Fit the response

Regress the response on the components by least squares.



repeat for M

Why this works

PLS chooses its directions with the response, so early components point where X predicts y .

PCR uses only X , so its directions capture the biggest spread in X , which need not help predict y .

Standardize first. A larger M keeps more detail; choose M by cross-validation.

